Multi-modal Data Spectrum

- Multi-modal Datasets are Multi-dimensional
- Madaan et al., ICLR 2026
- An audit of modality dependence across 23 VQA benchmarks

> 通过输入扰动把总体准确率拆成单模态依赖和跨模态交互。论文重点是评测诊断。

Paper v2 Code

A correct answer can use a shortcut

- Question priors can reveal answers when the image changes.
- Image content and answer options can reveal a likely label despite an unrelated question.
- Benchmark design that removes text shortcuts can accidentally strengthen image shortcuts.

> 多模态输入不保证模型真正联合使用了两种模态。这里的 shortcut 是基于实验观察的依赖，不是每题的因果证明。

Paper v2 Code

Four controlled evaluation conditions

Condition	Image	Question
Normal	Original	Original
Image contribution	Original	Shuffled
Text contribution	Shuffled	Original
Random baseline	Shuffled	Shuffled

> 论文使用随机替换而非直接删去模态。“Image only” 仍存在不相关文本，需明确这种实验操作。第 3 节。

Paper v2 Code

Dependency strength follows performance changes

- The adapted perceptual-score method compares normal and shuffled conditions.
- Single-modality performance above the random baseline exposes exploitable information.
- High paired performance with large drops under either shuffle suggests stronger interaction dependence.

> 这些量依赖选定模型和 metric。不能把它们当成数据集固有且唯一的信息论分解。

Paper v2 Code

Study coverage and model controls

- The study spans 23 multiple-choice VQA benchmarks.
- Domains include knowledge reasoning, spatial understanding, OCR, and charts.
- Experiments compare Cambrian scales and other MLLM families.

> 作者强调最好选择未在目标数据集训练过的 baseline，但现实 checkpoint 的污染与训练明细仍会限制解释。第 4.1 节。

Paper v2 Code

Predominantly interaction-dependent cases are rare

- The paper highlights MME, POPE, COCO, and V* as the clearest interaction-dependent cases.
- Several use complementary yes/no questions for the same images.
- A single aligned modality often remains informative on the other benchmark families.

> 这是该实验和模型下的分类，不应写成所有模型永远不能利用这些数据集的单模态偏差。第 4.2 节。

Paper v2 Code

Question-only gains above random

Benchmark	Reported absolute gain
GQA	26 points
ScienceQA	17.5 points
MMMU	11.35 points
AI2D	34.94 points

> 第 4.2 节描述的模型评测结果。gain 是相对 random 条件的百分点，不是相对正常多模态准确率提升。

Paper v2 Code

Dataset averages conceal category differences

- Dependence varies within a benchmark as well as across benchmarks.
- Larger models can improve total accuracy while preserving shortcut use.
- A benchmark label such as spatial reasoning does not establish image–question interaction for every sample.

> 选择评测集合时应按 category 分层看 shuffle gap，不能只按数据集名字做能力覆盖矩阵。

Paper v2 Code

Multiple-choice scope limits generalization

- Answer options can leak task structure and visually associated labels.
- Shuffling produces out-of-distribution inputs and can trigger behavior beyond information loss.
- The paper calls for open-ended evaluation and meaningful abstention.

> 论文没有证明所有开放式 caption 或 agent 任务存在同等偏差。需要重新定义输出与评分协议。

Paper v2 Code

Benchmark registry implications

- Inference: register input permutations, seeds, and answer-option formatting with every evaluation run.
- Proposed application: attach a modality-dependence profile to benchmark versions.
- Proposed application: compare category profiles before replacing one evaluation set with another.

> 这样才能区分模型进步与评测集合更换。生产 dashboard 应展示正常分数和三种对照。

Paper v2 Code

Code and reproducibility surface

- The public repository includes benchmark-specific generation and scoring paths.
- Normal, Text Shuffle, Image Shuffle, and Random runs produce separate JSONL outputs.
- Dataset and model downloads remain external dependencies with their own access and license constraints.

> 本次静态核查 README，未运行自动下载或推理。复现时需冻结 checkpoint、数据 revision 和随机种子。

Paper v2 Code

A practical acceptance test

- Run all four conditions on a stratified subset with fixed seeds.
- Compare per-category confidence, abstention, and correctness under altered modalities.
- Promote a benchmark only after its dependence profile matches the capability being claimed.

> 以上是从研究结论推导的评测流程，不能把 shuffle gap 的改善自动等同于业务任务改善。

Paper v2 Code